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AIR CLEANER

Abstract

Disclosed is an air cleaner. The air cleaner includes a case including a suction port disposed at a peripheral surface of the case, a first filter disposed in the case, the first filter being configured to remove foreign matter from air introduced into the suction port, a first blower fan disposed above the first filter, a first discharge port disposed above the first blower fan, a second filter disposed in the case, the second filter being configured to remove foreign matter from air introduced into the suction port, a second blower fan disposed below the second filter, a second discharge port disposed below the second blower fan, and a dual-shaft motor including shafts, the dual-shaft motor being disposed between the first blower fan and the second blower fan, and the shafts protruding from two different sides of the dual-shaft motor in a vertical direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims the priority benefit of Korean Patent Application No. 10-2024-0033177, filed on Mar. 8, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present disclosure relates to an air cleaner, and more particularly to an air cleaner capable of improving energy efficiency.
2. Description of the Related Art
[0003] An air cleaner is a device that filters air in a certain space and discharges the filtered air, thereby reducing dust or bacteria in the air in the corresponding space. The air cleaner filters out foreign matter by generating a flow of air in the corresponding space, and discharges air with the foreign matter removed therefrom.
[0004] Related Art Document 1 (Korean Patent Laid-Open Publication No. 10-2022-0083719) and Related Art Document 2 (Korean Patent Laid-Open Publication No. 10-2017-0140578) disclose air cleaners, in each of which blowers are disposed in a vertical direction to filter a large amount of air. Such an air cleaner of the related art has an advantage in that the blowing capacity is increased.
[0005] However, in the air cleaner of the related art, because each of the blowers is provided with a motor to rotate a fan, a large amount of energy is consumed to operate the fan, whereby power consumption may increase and energy efficiency may decrease. Further, because a space for mounting components of each of the blowers is necessary, the overall height of the air cleaner of the related art increases.
SUMMARY OF THE INVENTION
[0006] An aspect of the present disclosure is directed to providing an air cleaner capable of reducing power consumption and improving energy efficiency.
[0007] Another aspect of the present disclosure is directed to providing an air cleaner capable of improving suction capacity of air suctioned into the air cleaner.
[0008] A further aspect of the present disclosure is directed to providing an air cleaner capable of simplifying an internal component mounting structure and reducing the height of the product.
[0009] The aspects of the present disclosure are not limited to the aspects mentioned above, and other aspects not mentioned herein will be clearly understood by those skilled in the art from the following description.
[0010] An air cleaner according to an embodiment of the present disclosure is configured such that blower fans are rotated by a dual-shaft motor, thereby improving energy efficiency.
[0011] An air cleaner according to an embodiment of the present disclosure is configured such that an air cleaning module configured to discharge purified air downwardly and an air cleaning module configured to discharge purified air upwardly are stacked in an upward-downward direction, thereby exhibiting improved air cleaning performance.
[0012] An air cleaner according to an embodiment of the present disclosure includes a case having a suction port formed in the peripheral surface thereof, a first filter disposed in the case to remove foreign matter from air introduced into the suction port, a first blower fan disposed above the first filter, a first discharge port formed above the first blower fan, a second filter disposed in the case to remove foreign matter from air introduced into the suction port, a second blower fan disposed below the second filter, a second discharge port formed below the second blower fan, and a dual-

shaft motor including shafts protruding from two opposite sides thereof in an upward-downward direction. The dual-shaft motor is disposed between the first blower fan and the second blower fan, and the shafts of the dual-shaft motor include a first shaft extending downward to be connected to the first blower fan and a second shaft extending upward to be connected to the second blower fan.

[0013] The suction port may include a first suction port formed below the first discharge port and a second suction port formed above the second discharge port.

[0014] The air cleaner may further include a first grille disposed inside the first discharge port to guide air flowing upward from the first blower fan in a radially outward direction and a second grille disposed inside the second discharge port to guide air flowing downward from the second blower fan in the radially outward direction.

[0015] The first shaft may be connected to the first blower fan through a central cavity formed in the first grille, and the second shaft may be connected to the second blower fan through a central cavity formed in the second grille.

[0016] Each of the first grille and the second grille may include a plurality of grilles bent in an outward direction.

[0017] The plurality of grilles may have a concentric circle structure.

[0018] The plurality of grilles may have a length greater than an interval between the plurality of grilles.

[0019] Each of the first grille and the second grille may include a plurality of grilles disposed so as to be inclined at a predetermined angle.

[0020] The air cleaner may further include a motor fastening part configured to accommodate the dual-shaft motor, and a plurality of grilles may be disposed on the peripheral surface of the motor fastening part.

[0021] The motor fastening part may include a first motor fastening part disposed at a lower position and including a first grille portion corresponding to the first discharge port and a second motor fastening part disposed at an upper position and including a second grille portion corresponding to the second discharge port.

[0022] The first grille may be inserted into and mounted in the first motor fastening part, and the second grille may be inserted into and mounted in the second motor fastening part.

[0023] The first grille may be located below the first grille portion, and the second grille may be located above the second grille portion.

[0024] The first and second discharge ports may have shapes corresponding to the shapes of the first and second grille portions.

[0025] The case may include a first case having a semi-cylindrical shape and having the suction port and the first discharge port and a second case having a semi-cylindrical shape and having the suction port and the second discharge port.

[0026] The air cleaner may further include a top cover disposed on the top of the case, a control module disposed inside the top cover, and a display panel disposed on the top of the control module.

[0027] An air cleaner according to another embodiment of the present disclosure includes a case having a suction port formed in the peripheral surface thereof, a first filter disposed in the case to remove foreign matter from air introduced into the suction port, a first blower fan disposed above the first filter, a first discharge port formed below the first filter, a second filter disposed in the case to remove foreign matter from air introduced into the suction port, a second blower fan disposed below the second filter, a second discharge port formed above the second filter, and a dual-shaft motor including shafts protruding from two opposite sides thereof in an upward-downward direction. The dual-shaft motor is disposed between the first blower fan and the second blower fan, and the shafts of the dual-shaft motor include a first shaft extending downward to be connected to the first blower fan and a second shaft extending upward to be connected to the second blower fan.

[0028] The air cleaner according to the embodiment of the present disclosure may further include a

grille disposed inside the first discharge port to guide air flowing downward from the first blower fan in a radially outward direction.
[0029] The air cleaner according to the embodiment of the present disclosure may further include a grille disposed inside the second discharge port to guide air flowing upward from the second blower fan in the radially outward direction.
[0030] The air cleaner according to the embodiment of the present disclosure may further include a motor fastening part configured to accommodate the dual-shaft motor.
[0031] The first blower fan and the second blower fan may be disposed symmetrically to each other with respect to the dual-shaft motor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0032] The above and other objects, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
[0033] FIG. **1** is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure;
[0034] FIG. **2** is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure;
[0035] FIG. **3** is an exploded view of the air cleaner according to the embodiment of the present disclosure;
[0036] FIG. **4** is a view showing a dual-shaft motor according to an embodiment of the present disclosure;
[0037] FIG. **5** is a view showing an inner grille according to an embodiment of the present disclosure;
[0038] FIGS. **6** and **7** are views showing a first grille according to an embodiment of the present disclosure;
[0039] FIGS. **8** and **9** are views showing a second grille according to an embodiment of the present disclosure; and
[0040] FIG. **10** is a cross-sectional view of an air cleaner according to another embodiment of the present disclosure.
DETAILED DESCRIPTION
[0041] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The present disclosure may, however, be embodied in many different forms, and should not be construed as being limited to the embodiments set forth herein.
[0042] In the drawings, illustration of parts unrelated to the description is omitted to clearly and briefly describe the present disclosure, and the same or extremely similar components are denoted by the same reference numerals throughout the specification.
[0043] As used herein, the terms with which the names of components are suffixed, “module” and “unit”, are assigned to facilitate preparation of this specification, and are not intended to suggest unique meanings or functions. Accordingly, the terms “module” and “unit” may be used interchangeably.
[0044] It will be understood that although the terms “first”, “second”, etc., may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another component.
[0045] FIG. **1** is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure, FIG. **2** is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure, and FIG. **3** is an exploded view of the air cleaner

according to the embodiment of the present disclosure.

[0046] Referring to FIGS. **1** to **3**, a case **10** may have a suction port **10***a* formed in the peripheral surface thereof to suction air from various directions. Air may be suctioned from all directions with respect to a central line passing through the internal center of the case **10** in an upward-downward direction. The suction port **10***a* is provided in plural, and the plurality of suction ports **10***a* is spaced apart from each other in the circumferential direction of the case **10**. The plurality of suction ports **10***a* is evenly formed in the circumferential direction along the peripheral surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0047] In this specification, the upward-downward direction or the vertical direction is defined as an axial direction. The axial direction may correspond to the central axis-direction of blower fans **140** and **240**, which will be described later, i.e., the motor shaft direction of the fans. The radial direction or the horizontal direction may be understood as a direction perpendicular to the axial direction. The circumferential direction may be understood as a circumferential direction of an imaginary circle that is formed when rotating about the axial direction with a distance in the radial direction as a rotational radius.

[0048] The air cleaner **1** includes air cleaning modules **100** and **200** that generate airflows. The air cleaning modules **100** and **200** include blower fans **140** and **240**, which are disposed in the case **10**, and filters **120** and **220**, which remove foreign matter from air introduced into the suction port **10***a*, respectively.

[0049] The suction port **10***a* allows the inside of the case **10** and the outside to communicate with each other. The suction port **10***a* is provided in plural. The plurality of suction ports **10***a* may be formed in the peripheral surface of the case **10**.

[0050] The plurality of suction ports **10***a* is evenly formed in the circumferential direction along the outer circumferential surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0051] The plurality of suction ports **10***a* is formed in a stripe shape that is elongated in the upward-downward direction. Alternatively, the plurality of suction ports **10***a* may be formed through perforation in a circular or elliptical shape.

[0052] As described above, because the case **10** is formed in a cylindrical shape and the plurality of suction ports **10***a* is formed along the outer circumferential surface of the case **10**, the amount of air suctioned may increase.

[0053] The air introduced into the suction ports **10***a* may pass through the filters **120** and **220**. Each of the filters **120** and **220** may be formed in a cylindrical shape and may have a filter surface that filters air.

[0054] Referring to FIGS. **1** to **3**, the air cleaning modules **100** and **200** include a first air cleaning module **100** and a second air cleaning module **200** that are disposed in the upward-downward direction.

[0055] For example, the second air cleaning module **200** may be disposed on the first air cleaning module **100**. Because the first air cleaning module **100** is disposed at a lower portion of the air cleaner **1**, the first air cleaning module **100** may be referred to as a "lower air cleaning module" or a "lower module", and because the second air cleaning module **200** is disposed at an upper portion of the air cleaner **1**, the second air cleaning module **200** may be referred to as an "upper air cleaning module" or an "upper module".

[0056] The first air cleaning module **100** and the second air cleaning module **200** include blower fans **140** and **240** and filters **120** and **220** configured to remove foreign matter from air introduced into the suction ports **10***a*, respectively.

[0057] The first air cleaning module **100** includes a first blower fan **140** and a first filter **120** configured to remove foreign matter from air introduced into the suction ports.

[0058] The second air cleaning module **200** includes a second blower fan **240** and a second filter **220** configured to remove foreign matter from air introduced into the suction ports.

[0059] The case **10** may have a cylindrical shape. The case **10** may include a first case **11**, which has a semi-cylindrical shape and covers portions of the outer circumferential surfaces of the air cleaning modules **100** and **200**, and a second case **12**, which has a semi-cylindrical shape and covers the remaining portions of the outer circumferential surfaces of the air cleaning modules **100** and **200**.

[0060] Alternatively, the case **10** may have a shape of a truncated cone (i.e. a cone with the top cut off), and each of the first case **11** and the second case **12** may have a shape of a semi-truncated cone.

[0061] The first case **11** and the second case **12** may be coupled to each other to form the external appearance of the air cleaner **1**. Because the first case **11** covers the front surface of the air cleaner **1**, the first case **11** may be referred to as a "front case" or a "front cover". Because the second case **12** covers the rear surface of the air cleaner **1**, the second case **12** may be referred to as a "rear case" or a "rear cover".

[0062] The first case **11** includes an upper portion **11***a* corresponding to the second air cleaning module **200** and a lower portion **11***b* corresponding to the first air cleaning module **100**. The suction ports **10***a* are formed in the upper portion **11***a* and the lower portion **11***b*.

[0063] The second case **12** includes an upper portion **12***a* corresponding to the second air cleaning module **200** and a lower portion **12***b* corresponding to the first air cleaning module **100**. The suction ports **10***a* are formed in the upper portion **12***a* and the lower portion **12***b*.

[0064] The suction ports **10***a* may include first suction ports **10***ab* formed in the lower portions **11***b* and **12***b* and second suction ports **10***aa* formed in the upper portions **11***a* and **12***a*.

[0065] A central discharge portion **30** is formed between the first case **11** and the second case **12**. The air introduced into the suction ports **10***a* is purified by the filters **120** and **220**, and the purified air is discharged through the central discharge portion **30**.

[0066] In another embodiment, the case **10** may include an upper case (not shown) and a lower case (not shown). In this case, the central discharge portion **30** may be formed between the upper case and the lower case.

[0067] The upper case may include a first upper case, which has a semi-cylindrical shape and covers a portion of the outer circumferential surface of the second air cleaning module **200**, and a second upper case, which has a semi-cylindrical shape and covers the remaining portion of the outer circumferential surface of the second air cleaning module **200**. In addition, the lower case may include a first lower case, which has a semi-cylindrical shape and covers a portion of the outer circumferential surface of the first air cleaning module **100**, and a second lower case, which has a semi-cylindrical shape and covers the remaining portion of the outer circumferential surface of the first air cleaning module **100**.

[0068] The first blower fan **140** is disposed above the first filter **120**. First discharge ports **30***a* are located above the first blower fan **140**. When the first blower fan **140** rotates, the air suctioned through the suction ports **10***a* formed in the lower cases **11***b* and **12***b* flows upward via the first filter **120**, and then is discharged through the first discharge ports **30***a*.

[0069] The second blower fan **240** is disposed below the second filter **220**. Second discharge ports **30***b* are located below the second blower fan **240**. The air suctioned through the suction ports **10***a* formed in the upper cases **11***a* and **12***a* flows downward via the second filter **220**, and then is discharged through the second discharge ports **30***b*.

[0070] The first discharge ports **30***a* and the second discharge ports **30***b* may be formed adjacent to the central portion of the air cleaner **1**. A combination of the first discharge ports **30***a* and the second discharge ports **30***b* may be referred to as the central discharge portion **30**.

[0071] The first suction ports **10***ab* may be located below the first discharge ports **30***a*, and the second suction ports **10***aa* may be located above the second discharge ports **30***b*. The lower suction airflow introduced into the first suction ports **10***ab* passes through the first filter **120**, and then is discharged through the first discharge ports **30***a* located above the first filter **120**. The upper suction airflow introduced into the second suction ports **10***aa* passes through the second filter **220**, and then is discharged through the second discharge ports **30***b* located below the second filter **220**.

[0072] A dual-shaft motor **500** with shafts protruding from two opposite sides thereof in the upward-downward direction is disposed between the first blower fan **140** and the second blower fan **240**.

[0073] FIG. **4** is a view showing a dual-shaft motor according to an embodiment of the present disclosure.

[0074] Referring to FIGS. **1** to **4**, the dual-shaft motor **500** includes a motor body **510** and shafts **515** protruding from two opposite sides of the motor body **510** in the upward-downward direction. The shafts **515** include a first shaft **515***a* extending downward to be connected to the first blower fan **140** and a second shaft **515***b* extending upward to be connected to the second blower fan **240**.

[0075] The first blower fan **140** is rotatably connected to the first shaft **515***a*. The second blower fan **240** is rotatably connected to the second shaft **515***b*.

[0076] A first gear **520** may be connected to the first shaft **515***a* and may be inserted into the central portion of the first blower fan **140** to be connected thereto. A second gear **530** may be connected to the second shaft **515***b* and may be inserted into the central portion of the second blower fan **240** to be connected thereto.

[0077] The first blower fan **140** and the second blower fan **240** may be rotated by rotational force received from the dual-shaft motor **500**. When the dual-shaft motor **500** operates, the first blower fan **140** and the second blower fan **240** connected to the dual-shaft motor **500** rotate at the same speed.

[0078] According to the present disclosure, because the two fans **140** and **240** are rotated by the single dual-shaft motor **500**, power consumption may be reduced, and energy efficiency may be improved compared to when two motors are used.

[0079] According to the present disclosure, the discharge portion **30** may be integrated with the central portion of the product, the discharge structure may be simplified, and an airflow may be formed intensively in the central portion of the air cleaner **1**.

[0080] Inner grilles **110** and **210** configured to guide air in the radially outward direction are disposed inside the central discharge portion **30**.

[0081] A first grille **110** is disposed inside the first discharge ports **30***a* in order to guide air flowing upward from the first blower fan **140** in the radially outward direction. A second grille **210** is disposed inside the second discharge ports **30***b* in order to guide air flowing downward from the second blower fan **240** in the radially outward direction. FIG. **5** is a view showing an inner grille according to an embodiment of the present disclosure, FIGS. **6** and **7** are views showing a first grille according to an embodiment of the present disclosure, and FIGS. **8** and **9** are views showing a second grille according to an embodiment of the present disclosure.

[0082] Referring to FIGS. **5** to **9**, each of the inner grilles **110** and **210** may include a plurality of grilles **613** having a concentric circle structure. The plurality of grilles **613** may be a plurality of circles having the same center and different radii. Each of the inner grilles **110** and **210** may further include an outer wall **612** and an inner wall **611**, between which the plurality of grilles **613** having a concentric circle structure is disposed.

[0083] A central cavity **614** may be located inside the inner wall **611**. The first shaft **515***a* of the dual-shaft motor **500** extends through the central cavity **614** in the first grille **110** to be connected to the first blower fan **140**. The second shaft **515***b* of the dual-shaft motor **500** extends through the central cavity **614** in the second grille **210** to be connected to the second blower fan **240**.

[0084] The lengths of the grilles **613** may be greater than an interval d**1** between the grilles **613**. Accordingly, it is possible to secure a sufficient distance for guiding an airflow in the lateral direction.

[0085] Each of the plurality of grilles **613** may include a section bent in the radially outward direction. Each of the inner grilles **110** and **210** may include a plurality of grilles **613** bent at a predetermined angle in the outward direction. The first grille **110** may be disposed inside the first discharge ports **30***a* in order to deliver the discharge airflow flowing upward in the lateral direction. The second grille **210** may be disposed inside the second discharge ports **30***b* in order to deliver the discharge airflow flowing downward in the lateral direction.

[0086] Each of the grilles **900** of the inner grilles **110** and **210** may be bent at a predetermined angle ag. The grille bending angle ag may be set to 20 degrees to 40 degrees. If the grille bending angle ag is less than 20 degrees, the airflow may not be sufficiently delivered in the lateral direction. If the grille bending angle ag is greater than **40** degrees, airflow loss may increase. The thickness d**2** of the grille **900** may not be constant in some sections thereof.

[0087] In another embodiment, each of the inner grilles **110** and **210** may include a plurality of linear grilles disposed so as to be inclined at a predetermined angle. The linear grilles may be disposed so as to be inclined at an angle corresponding to the discharge angle ag. Also in this case, the airflow may be formed along the discharge angle ag corresponding to the angle at which the linear grilles are disposed.

[0088] The dual-shaft motor **500** may be accommodated in motor fastening parts **125** and **225**. A plurality of grilles may be disposed on the peripheral surfaces of the motor fastening parts **125** and **225**.

[0089] The motor fastening parts **125** and **225** include a first motor fastening part **125** disposed at a lower position and a second motor fastening part **225** disposed at an upper position. The first motor fastening part **125** and the second motor fastening part **225** may be coupled to each other to define an inner space in which the dual-shaft motor **500** is accommodated.

[0090] The first motor fastening part **125** includes a first grille portion **125***a* corresponding to the first discharge ports **30***a*, and the second motor fastening part **225** includes a second grille portion **225***a* corresponding to the second discharge ports **30***b*.

[0091] The first grille **110** may be inserted into and mounted in the first motor fastening part **125**, and the second grille **210** may be inserted into and mounted in the second motor fastening part **225**. The first motor fastening part **125** may further include a coupling portion **125***b* for accommodation of the dual-shaft motor **500** and mounting of the first grille **110**. The second motor fastening part **225** may further include a coupling portion **225***b* for accommodation of the dual-shaft motor **500** and mounting of the second grille **210**.

[0092] In this case, the first grille **110** may be located below the first grille portion **125***a*, and the second grille **210** may be located above the second grille portion **225***a*.

[0093] In some embodiments, the shapes of the first and second discharge ports **30***a* and **30***b* may correspond to the shapes of the first and second grille portions **125***a* and **225***a*. For example, if the first and second discharge ports **30***a* and **30***b* are formed in a stripe type, and the first and second grille portions **125***a* and **225***a* may also be formed in a stripe type.

[0094] According to the present disclosure, because the dual-shaft motor **500** and the motor fastening parts **125** and **225**, which are commonly used, are disposed between the first and second air cleaning modules **100** and **200**, the height of the product may be reduced, and manufacturing costs may be reduced.

[0095] The first air cleaning module **100** includes a first filter mounting part **160** to which the first filter **120** is mounted. The first filter **120** may be removably mounted to the first filter mounting part **160**. The first filter **120** may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter **120**. While the air passes through the first filter **120**, impurities such as fine dust may be removed from the air.

[0096] Because the first filter **120** has a cylindrical shape, air is capable of being introduced into the first filter **120** from any direction. Therefore, the air filtering area may increase.

[0097] The first filter mounting part **160** may be formed in a cylindrical shape corresponding to the shape of the first filter **120**. In the process of mounting the first filter **120**, the first filter **120** may be slidably fitted into the first filter mounting part **160**. Conversely, in the process of demounting the first filter **120**, the first filter **120** may be slidably withdrawn from the first filter mounting part **160**.

[0098] The first air cleaning module **100** includes a first fan housing **145** disposed above the first filter **120**. The first blower fan **140** may be rotatably disposed in the first fan housing **145**.

[0099] The first blower fan **140** suctions air in the axial direction and discharges the air upwardly in the radial direction. The first fan housing **145** may form a fan path that accommodates the first blower fan **140** and guides the air moved by the first blower fan **140** upwardly.

[0100] The second air cleaning module **200** also includes a second filter mounting part **260** to which the second filter **220** is mounted. The above description of the first filter **120** and the first filter mounting part **160** may be similarly applied to the second filter **220** and the second filter mounting part **260**. However, while the first filter **120** and the first filter mounting part **160** are located below the first blower fan **140** and the first fan housing **145**, the second filter **220** and the second filter mounting part **260** are located above the second blower fan **240** and the second fan housing **245**.

[0101] The second blower fan **240** suctions air in the axial direction and discharges the air downwardly in the radial direction. The second fan housing **245** may form a fan path that accommodates the second blower fan **240** and guides the air moved by the second blower fan **240** downwardly.

[0102] When the first filter **120** is mounted to the first filter mounting part **160** of the first air cleaning module **100**, the first filter **120** presses a lower filter pressing plate **350**. When the second filter **220** is mounted to the second filter mounting part **260** of the second air cleaning module **200**, the second filter **220** presses an upper filter pressing plate **340**. The first air cleaning module **100** may include a lower filter mounting part cover **320**, and the second air cleaning module **200** may include an upper filter mounting part cover **310**.

[0103] A support **170** is disposed below the first air cleaning module **100**. The support **170** may be coupled to a plate **330**. The support **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0104] A top cover **270** is disposed on the top of the case **10**. A control module **400** is disposed inside the top cover **270**. The control module **400** may accommodate a main printed circuit board (PCB) on which circuits such as a controller that controls overall operation of the air cleaner **1** are mounted.

[0105] A display **390** is disposed on the top of the control module **400**. In addition, an operation button for operation of the air cleaner **1** may be further disposed on the top of the control module **400**. A display base **350** may support the display **390** from below.

[0106] According to the present disclosure, because the dual-shaft motor **500** is disposed at the center of the air cleaner **1**, the upper and lower blower fans **140** and **240** are capable of being rotated by the single motor, whereby the amount of power consumed during rated operation may be reduced, and energy efficiency may be improved. Further, because the number of motors is reduced, the component fastening structure may be simplified, and the overall height of the product may be reduced.

[0107] Unlike the above-described embodiment, the discharge ports may be divided into upper discharge ports disposed at an upper portion of the air cleaner **1** and lower discharge ports disposed at a lower portion of the air cleaner **1**.

[0108] FIG. **10** is a cross-sectional view of an air cleaner according to another embodiment of the present disclosure, in which discharge ports are divided into upper discharge ports disposed at an upper portion of the air cleaner **1** and lower discharge ports disposed at a lower portion of the air cleaner **1**.

[0109] Hereinafter, another embodiment will be described with reference to FIG. **10**. The following description will focus on differences from the embodiment described above with reference to FIGS. **1** to **9**. With regard to any non-described component of the other embodiment shown in FIG. **10**, reference may be made to the description of the above-described embodiment.

[0110] Referring to FIG. **10**, the first air cleaning module **100** includes a first blower fan **140***a* and a first filter **120** that removes foreign matter from the air introduced into the suction ports. First discharge ports **110***b* may be disposed below the first filter **120**.

[0111] The second air cleaning module **200** includes a second blower fan **240***a* and a second filter **220** that removes foreign matter from the air introduced into the suction ports. Second discharge ports **210***b* may be disposed above the second filter **220**.

[0112] A dual-shaft motor **500** with shafts protruding from two opposite sides thereof in the upward-downward direction is disposed between the first blower fan **140***a* and the second blower fan **240***a*. The dual-shaft motor **500** may be accommodated in motor fastening parts **125** and **225**. Because the discharge ports are not formed at the center of the product, a plurality of grilles does not need to be disposed on the peripheral surfaces of the motor fastening parts **125** and **225**.

[0113] When the dual-shaft motor **500** operates, the first blower fan **140***a* and the second blower fan **240***a* connected to the dual-shaft motor **500** rotate. The air introduced into the suction ports **10***a* formed in the side surface of the case **10** may pass through the filters **120** and **220**.

[0114] The first blower fan **140***a* and the second blower fan **240***a* may be disposed symmetrically to each other with respect to the dual-shaft motor **500**.

[0115] The first blower fan **140***a* shown in FIG. **10** may be disposed so as to be oriented opposite the first blower fan **140** shown in FIG. **2** in the upward-downward direction, thereby blowing the air downwardly. As the first blower fan **140***a* rotates, the air suctioned through the suction ports **10***a* flows downward via the first filter **120**, and then is discharged through the first discharge ports **110***b*. A grille **110***a* may be disposed inside the first discharge ports **110***b* in order to deliver the discharge airflow flowing downward in the lateral direction. For example, the grille **110***a* may be implemented in the same shape as the grille **210** illustrated in FIG. **8**.

[0116] The second blower fan **240***a* shown in FIG. **10** may be disposed so as to be oriented opposite the second blower fan **240** shown in FIG. **2** in the upward-downward direction, thereby blowing the air upwardly. As the second blower fan **240***a* rotates, the air suctioned through the suction ports **10***a* flows upward via the second filter **220**, and then is discharged through the second discharge ports **210***b*. A grille **210***a* may be disposed inside the second discharge ports **210***b* in order to deliver the discharge airflow flowing upward in the lateral direction. For example, the grille **210***a* may be implemented in the same shape as the grille **110** illustrated in FIG. **6**.

[0117] As is apparent from the above description, according to at least one of the embodiments of the present disclosure, a first blower fan is disposed at an upper position, a second blower fan is disposed at a lower position, and a dual-shaft motor is disposed so that shafts protruding from two opposite sides thereof in the upward-downward direction are connected to the first and second blower fans, whereby power consumption may be reduced, and energy efficiency may be improved.

[0118] Further, because the dual-shaft motor and motor fastening parts, which are commonly used, are disposed at the center of the air cleaner, the height of the product may be reduced, and manufacturing costs may be reduced.

[0119] Furthermore, first discharge ports are formed above the first blower fan, second discharge ports are formed below the second blower fan, first suction ports are formed below the first discharge ports, and second suction ports are formed above the second discharge ports, whereby suction airflows may be formed in the upper and lower portions of the product, and discharge airflows may be formed intensively in the central portion of the product.

[0120] Furthermore, a first grille is disposed inside the first discharge ports in order to guide air flowing upward from the first blower fan in the radially outward direction, and a second grille is disposed inside the second discharge ports in order to guide air flowing downward from the second blower fan in the radially outward direction, whereby the discharge direction of air may be effectively guided.

[0121] The effects achievable through the disclosure are not limited to the above-mentioned effects,

and other effects not mentioned herein will be clearly understood by those skilled in the art from the description of the claims.
[0122] Other effects will be clearly understood by those skilled in the art from the description of the claims.
[0123] Although the present disclosure has been described with reference to specific embodiments shown in the drawings, it is apparent to those skilled in the art that the present disclosure is not limited to those embodiments and is embodied in many forms without departing from the scope of the present disclosure, which is described in the following claims. These modifications should not be individually understood from the technical spirit or scope of the present disclosure.

Claims

1. An air cleaner comprising: a case comprising a suction port disposed at a peripheral surface of the case; a first filter disposed in the case, the first filter being configured to remove foreign matter from air introduced into the suction port; a first blower fan disposed above the first filter; a first discharge port disposed above the first blower fan; a second filter disposed in the case, the second filter being configured to remove foreign matter from air introduced into the suction port; a second blower fan disposed below the second filter; a second discharge port disposed below the second blower fan; and a dual-shaft motor comprising shafts, the dual-shaft motor being disposed between the first blower fan and the second blower fan, and the shafts protruding from opposite sides of the dual-shaft motor in a vertical direction, the shafts comprising: a first shaft extending downward and connected to the first blower fan; and a second shaft extending upward and connected to the second blower fan.
2. The air cleaner according to claim 1, wherein the suction port comprises: a first suction port disposed below the first discharge port; and a second suction port disposed above the second discharge port.
3. The air cleaner according to claim 2, further comprising: a first grille disposed in the first discharge port, the first grille being configured to guide air flowing upward from the first blower fan in a radially outward direction with respect to the air cleaner; and a second grille disposed in the second discharge port, the second grille being configured to guide air flowing downward from the second blower fan in the radially outward direction with respect to the air cleaner.
4. The air cleaner according to claim 3, wherein the first shaft is connected to the first blower fan through a central cavity disposed in the first grille, and wherein the second shaft is connected to the second blower fan through a central cavity disposed in the second grille.
5. The air cleaner according to claim 3, wherein each of the first grille and the second grille comprises a plurality of grilles bent in an outward direction with respect to the air cleaner.
6. The air cleaner according to claim 5, wherein the plurality of grilles are arranged in a concentric circle structure.
7. The air cleaner according to claim 5, wherein each grille of the plurality of grilles has a length greater than a length of an interval between adjacent grilles of the plurality of grilles.
8. The air cleaner according to claim 3, wherein each of the first grille and the second grille comprises a plurality of grilles, and wherein each grille of the plurality of grilles is inclined at an angle with respect to the vertical direction.
9. The air cleaner according to claim 3, further comprising a motor fastening part configured to accommodate the dual-shaft motor, wherein a plurality of grilles are disposed on a peripheral surface of the motor fastening part.
10. The air cleaner according to claim 9, wherein the motor fastening part comprises: a first motor fastening part disposed at a lower position of the motor fastening part, the first motor fastening part comprising a first grille portion corresponding to the first discharge port; and a second motor fastening part disposed at an upper position of the motor fastening part, the second motor fastening

part comprising a second grille portion corresponding to the second discharge port.

11. The air cleaner according to claim 10, wherein the first grille portion is configured to surround a first portion of the dual-shaft motor, and wherein the second grille portion is configured to surround a second portion of the dual-shaft motor.

12. The air cleaner according to claim 10, wherein the first grille is located below the first grille portion, and wherein the second grille is located above the second grille portion.

13. The air cleaner according to claim 10, wherein the first discharge port and the second discharge port have shapes corresponding to shapes of the first grille portion and the second grille portion, respectively.

14. The air cleaner according to claim 1, wherein the case further comprises: a first case having a semi-cylindrical shape, the first case including the first discharge port and a first portion of the suction port; and a second case having a semi-cylindrical shape, the second case including the second discharge port and a second portion of the suction port.

15. The air cleaner according to claim 1, further comprising: a top cover disposed on a top of the case; a control module disposed inside the top cover; and a display panel disposed on a top of the control module.

16. An air cleaner comprising: a case including a suction port disposed at a peripheral surface of the case; a first filter disposed in the case, the first filter being configured to remove foreign matter from air introduced into the suction port; a first blower fan disposed above the first filter; a first discharge port disposed below the first filter; a second filter disposed in the case, the second filter being configured to remove foreign matter from air introduced into the suction port; a second blower fan disposed below the second filter; a second discharge port disposed above the second filter; and a dual-shaft motor comprising shafts, the dual-shaft motor being disposed between the first blower fan and the second blower fan, and the shafts protruding from opposite sides of the dual-shaft motor in a vertical direction, the shafts comprising: a first shaft extending downward and connected to the first blower fan; and a second shaft extending upward and connected to the second blower fan.

17. The air cleaner according to claim 16, further comprising: a first grille disposed in the first discharge port, the first grille being configured to guide air flowing downward from the first blower fan in a radially outward direction with respect to the air cleaner; and a second grille disposed in the second discharge port, the second grille being configured to guide air flowing upward from the second blower fan in the radially outward direction with respect to the air cleaner.

18. An air cleaner comprising: a case including a suction port disposed at a peripheral surface of the case; a first cleaning module including a first filter configured to remove foreign matter from air introduced into the suction port, a first blower fan and a first discharge port; a second cleaning module disposed above the first cleaning module and including a second filter configured to remove foreign matter from air introduced into the suction port, a second blower fan and a second discharge port; and a dual-shaft motor including a portion that overlaps the first discharge port and the second discharge port with respect to a vertical direction, the dual-shaft motor comprising: a motor body; a first shaft protruding from a first side of the motor body, the first shaft being configured to rotate the first blower fan; and a second shaft protruding from a second side of the motor body, the second shaft being configured to rotate the second blower fan.

19. The air cleaner according to claim 18, wherein the dual-shaft motor further comprises: a first gear disposed on the first shaft, the first gear being coupled to the first blower fan; and a second gear disposed on the second shaft, the second gear being coupled to the second blower fan.

20. The air cleaner according to claim 19, wherein the first blower fan and the second blower fan each include a central cavity, wherein the first gear is coupled to the first blower fan from inside the central cavity of the first blower fan, and wherein the second gear is coupled to the second blower fan from inside the central cavity of the second blower fan.